



Session 16B: Advanced Integration — Parts, Trig Integrals, Trig Sub, Partial Fractions

Phase 2 — Classical Techniques | 85 min

Four technique families. Each has a clear trigger, a step-by-step procedure, and a telltale pattern. The hard part isn't executing any single technique — it's choosing the right one. This session teaches you the decision tree.

Prerequisites: 16A (FTC & u -sub), 14B (chain/product rules), 11A (trig)

The Integration Decision Tree

Before you start, run this checklist top to bottom. The first match wins.

Look at the integral. Ask:

- | | |
|--|--------------------------------------|
| 1. Is it a basic form from the dictionary? | → Antiderivative dictionary (16A) |
| 2. Is it $f(g(x)) \cdot g'(x)$ — a function AND its deriv? | → u -substitution (16A) |
| 3. Is it a PRODUCT of two different function types? | → Integration by PARTS (16B-A) |
| 4. Is it $\sin^m \cos^n$ or $\tan^m \sec^n$? | → Trigonometric INTEGRALS (16B-B) |
| 5. Is there $\sqrt{a^2 \pm x^2}$ or $\sqrt{x^2 - a^2}$? | → Trigonometric SUBSTITUTION (16B-C) |
| 6. Is it a rational function $P(x)/Q(x)$? | → PARTIAL FRACTIONS (16B-D) |

Part A: Integration by Parts — Undoing the Product Rule

Trigger: A product of two unrelated function families (polynomial $\times$ exponential, log $\times$ polynomial, inverse trig $\times$ anything).

Procedure: $\int u dv = uv - \int v du$. Choose u using LIATE. Let dv be everything else.

Example 1: The Parts Algorithm — Six Steps

Algorithm for $\int f(x)g(x) dx$:

Step	Action
1	Choose u using LIATE: L og, I nverse trig, A lgebraic (x^n), T rig, E xponential
2	Set dv = everything else (including dx)
3	Compute du = derivative of u , times dx
4	Compute v = integrate dv (no $+C$ needed here)
5	Assemble: $uv - \int v du$
6	Integrate $\int v du$. It should be simpler than the original

Example 2: Polynomial $\times$ Exponential — LIATE in Action

$\int xe^x dx$.

Step	Action
1	$u = x$ (Algebraic beats Exponential)
2	$dv = e^x dx$
3	$du = 1 \cdot dx$
4	$v = \int e^x dx = e^x$
5	$uv - \int v du = xe^x - \int e^x dx$

Step	Action
6	$= xe^x - e^x + C = e^x(x - 1) + C$

Check: $\frac{d}{dx}[e^x(x - 1)] = e^x(x - 1) + e^x(1) = xe^x. \checkmark$

Example 3: Log × Anything — Log Always Wins (LIATE Priority 1)

$\int \ln x \, dx$. This is a product of $\ln x$ and 1 (implicit).

Step	Action
1	$u = \ln x$ (Log — highest priority)
2	$dv = 1 \cdot dx$
3	$du = \frac{1}{x} dx$
4	$v = \int 1 \, dx = x$
5, 6	$x \ln x - \int x \cdot \frac{1}{x} \, dx = x \ln x - \int 1 \, dx = x \ln x - x + C$

Memorize: $\int \ln x \, dx = x \ln x - x + C$. This comes up constantly.

Example 4: Inverse Trig — LIATE Priority 2

$\int \arctan x \, dx$.

Step	Action
1	$u = \arctan x$ (Inverse trig beats implicit 1)
2	$dv = 1 \cdot dx$

Step	Action
3	$du = \frac{1}{1+x^2} dx$
4	$v = x$
5, 6	$x \arctan x - \int \frac{x}{1+x^2} dx$

The new integral is now a u -sub: $w = 1 + x^2$, $dw = 2x dx$.
 $= x \arctan x - \frac{1}{2} \ln(1 + x^2) + C$.

Example 5: Two Rounds — Trig × Exponential (The Cycling Pattern)

$\int e^x \sin x dx$. Neither function simplifies when differentiated. The integral **cycles**.

Procedure for cycling integrals:

1. Apply parts twice. Name the original integral I .
2. When I reappears, solve for it algebraically.
3. Add $+C$ at the very end.

Round	u	dv	du	v	Result
1	$\sin x$	$e^x dx$	$\cos x dx$	e^x	$I = e^x \sin x - \int e^x \cos x dx$
2	$\cos x$	$e^x dx$	$-\sin x dx$	e^x	$I = e^x \sin x - (e^x \cos x + I)$

Simplify: $I = e^x \sin x - e^x \cos x - I \rightarrow 2I = e^x(\sin x - \cos x) \rightarrow I = \frac{e^x}{2}(\sin x - \cos x) + C$.

Add $+C$ only when the integral is fully solved — at the end.

Example 6: Parts Quick Reference

If you see...	Choose $u =$	Why
$x^n e^{kx}$	$u = x^n$	Differentiating polynomial lowers its degree → simpler
$x^n \sin x$ or $x^n \cos x$	$u = x^n$	Same — polynomial simplifies
$(\ln x) \times (\text{anything})$	$u = \ln x$	Derivative is $\frac{1}{x}$, which tames the rest
$(\arctan x) \times$ (anything)	$u =$ inverse trig	Derivative is algebraic → simpler
$e^{kx} \sin(mx)$ or $e^{kx} \cos(mx)$	Either	It cycles — apply twice, solve for I

Part B: Trigonometric Integrals — Exploiting $\sin^2 + \cos^2 = 1$

Trigger: The integrand is built from $\sin^m x$, $\cos^n x$, $\tan^m x$, $\sec^n x$.

Strategy: Use identities to reduce to a u -sub.

Example 7: $\int \sin^m x \cos^n x dx$ — The Parity Decision Tree

Check the exponents m and n :

Is m ODD?

- Peel off ONE $\sin x$.
- Convert $\sin^{m-1} x$ to $(1-\cos^2 x)^{(m-1)/2}$.
- $u = \cos x$, $du = -\sin x dx$.

Is n ODD? (if m is even)

- Peel off ONE $\cos x$.
- Convert $\cos^{n-1} x$ to $(1-\sin^2 x)^{(n-1)/2}$.
- $u = \sin x$, $du = \cos x dx$.

Are BOTH EVEN?

- Use half-angle: $\sin^2 x = (1-\cos 2x)/2$, $\cos^2 x = (1+\cos 2x)/2$.
- Reduces powers. Repeat if needed.

Example 8: Odd Power — Peel, Convert, u -Sub

$\int \sin^3 x dx$. $m = 3$ (odd). Follow the "m odd" branch.

1. **Peel:** $\sin^3 x = \sin x \cdot \sin^2 x$.
 2. **Convert:** $\sin^2 x = 1 - \cos^2 x$. Integral: $\int \sin x (1 - \cos^2 x) dx$.
 3. **u -sub:** $u = \cos x$, $du = -\sin x dx \rightarrow \sin x dx = -du$.
 4. **Integrate:** $\int (1 - u^2)(-du) = -u + \frac{u^3}{3} + C$.
 5. **Back:** $-\cos x + \frac{\cos^3 x}{3} + C$.
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Example 9: Even Power — Half-Angle Formula

$\int \cos^2 x \, dx$. $n = 2$ (even, and $m = 0$ which is even too).

1. **Half-angle:** $\cos^2 x = \frac{1+\cos 2x}{2}$.
 2. **Integrate:** $\int \frac{1+\cos 2x}{2} \, dx = \frac{1}{2} \int (1 + \cos 2x) \, dx$.
 3. $= \frac{1}{2} \left(x + \frac{\sin 2x}{2} \right) + C = \frac{x}{2} + \frac{\sin 2x}{4} + C$.
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Example 10: $\int \tan^m x \sec^n x \, dx$ — Save $\sec^2 x$

Is n EVEN ($n \geq 2$)?

- Save ONE $\sec^2 x$. It becomes du .
- Convert remaining $\sec^{n-2}$ to $(\tan^2 x + 1)^{(n-2)/2}$.
- $u = \tan x$, $du = \sec^2 x \, dx$.

Is m ODD ($m \geq 1$)?

- Save ONE $\sec x \tan x$. It becomes du .
- Convert remaining $\tan^{m-1}$ to $(\sec^2 x - 1)^{(m-1)/2}$.
- $u = \sec x$, $du = \sec x \tan x \, dx$.

Otherwise → convert everything to $\sin/\cos$ and use the $\sin^m \cos^n$ strategy.

$\int \tan^3 x \sec^2 x \, dx$: $n = 2$ (even). Save $\sec^2 x$ for du .

$u = \tan x$, $du = \sec^2 x \, dx$. Integral $= \int u^3 \, du = \frac{\tan^4 x}{4} + C$.

Part C: Trigonometric Substitution — Killing the Square Root

Trigger: The integrand contains $\sqrt{a^2 \pm x^2}$ or $\sqrt{x^2 - a^2}$.

Strategy: Replace x with a trig function. The root disappears via a Pythagorean identity.

Example 11: The Trig Sub Lookup Table

Form	Substitute	dx becomes	The root simplifies to
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$	$a \cos \theta d\theta$	$a \cos \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$	$a \sec^2 \theta d\theta$	$a \sec \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$	$a \sec \theta \tan \theta d\theta$	$a \tan \theta$

Example 12: Trig Sub Step by Step

Step	Action	For $\int \frac{dx}{\sqrt{4-x^2}}$
1	Identify the root form	$\sqrt{4-x^2} = \sqrt{2^2-x^2} \rightarrow$ Form 1
2	Substitute x and dx	$x = 2 \sin \theta, dx = 2 \cos \theta d\theta$
3	Simplify the root	$\sqrt{4-x^2} = \sqrt{4-4\sin^2 \theta} = 2 \cos \theta$
4	Rewrite integral in θ	$\int \frac{2 \cos \theta d\theta}{2 \cos \theta} = \int d\theta$
5	Integrate	$\theta + C$
6	Back-substitute via right triangle	$\theta = \arcsin(x/2)$

Example 13: $\sqrt{a^2 + x^2}$ — The Tangent Case

$\int \frac{dx}{\sqrt{x^2+1}}$. Form: $\sqrt{x^2+1} \rightarrow x = \tan \theta, a = 1$.

- $x = \tan \theta, dx = \sec^2 \theta d\theta$.
- $\sqrt{x^2+1} = \sqrt{\tan^2 \theta + 1} = \sec \theta$.
- Integral: $\int \frac{\sec^2 \theta d\theta}{\sec \theta} = \int \sec \theta d\theta$.

4. $\int \sec \theta \, d\theta = \ln |\sec \theta + \tan \theta| + C$. (Standard — memorize.)
5. **Back-substitute:** Draw a right triangle with angle θ , opposite = x , adjacent = 1. Hypotenuse = $\sqrt{x^2 + 1}$. So $\sec \theta = \sqrt{x^2 + 1}$, $\tan \theta = x$.
6. Answer: $\ln |x + \sqrt{x^2 + 1}| + C$.

Example 14: The Right Triangle Method

After integrating in θ , draw a right triangle to express trig functions in x :

Substitution	Triangle label	$\sin \theta$	$\cos \theta$	$\tan \theta$	$\sec \theta$
$x = a \sin \theta$	Opp= x , Hyp= a	x/a	$\sqrt{a^2 - x^2}/a$	$x/\sqrt{a^2 - x^2}$	—
$x = a \tan \theta$	Opp= x , Adj= a	$x/\sqrt{x^2 + a^2}$	$a/\sqrt{x^2 + a^2}$	x/a	$\sqrt{x^2 + a^2}/a$
$x = a \sec \theta$	Hyp= x , Adj= a	$\sqrt{x^2 - a^2}/x$	a/x	$\sqrt{x^2 - a^2}/a$	x/a

Part D: Partial Fractions — Splitting Rational Functions

Trigger: The integrand is $\frac{P(x)}{Q(x)}$ — a rational function with a factorable denominator.

Strategy: Decompose into a sum of simpler fractions, each matching the dictionary.

Example 15: The Partial Fractions Algorithm

Step	Action
1	Check degrees: If $\deg(P) \geq \deg(Q)$, do polynomial long division FIRST
2	Factor $Q(x)$ into linear factors $(ax + b)^k$ and irreducible quadratics $(ax^2 + bx + c)^m$
3	Write decomposition using the template below
4	Solve for constants $A, B, C, \dots$ (plug roots or match coefficients)
5	Integrate each term

Example 16: Decomposition Templates

Factor type	Template
$(ax + b)$ — distinct linear	$\frac{A}{ax + b}$
$(ax + b)^k$ — repeated linear	$\frac{A_1}{ax + b} + \frac{A_2}{(ax + b)^2} + \dots + \frac{A_k}{(ax + b)^k}$
$ax^2 + bx + c$ — irreducible quadratic	$\frac{Ax + B}{ax^2 + bx + c}$

Example 17: Distinct Linear Factors

$$\int \frac{1}{x^2-1} dx = \int \frac{1}{(x-1)(x+1)} dx.$$

1. Degree: $0 < 2$. No division.
2. Denominator already factored: $(x - 1)(x + 1)$.
3. Template: $\frac{1}{(x-1)(x+1)} = \frac{A}{x-1} + \frac{B}{x+1}$.

4. Clear denominators: $1 = A(x + 1) + B(x - 1)$.

Solve by plugging roots (fastest):

- Plug $x = 1$: $1 = 2A \rightarrow A = 1/2$.
- Plug $x = -1$: $1 = -2B \rightarrow B = -1/2$.

5. Integrate: $\int \left(\frac{1/2}{x-1} - \frac{1/2}{x+1} \right) dx = \frac{1}{2} \ln |x - 1| - \frac{1}{2} \ln |x + 1| + C = \frac{1}{2} \ln \left| \frac{x-1}{x+1} \right| + C$.

Example 18: Repeated Linear Factor

$$\int \frac{x}{(x-1)^2} dx.$$

Template: $\frac{x}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$.

Clear: $x = A(x - 1) + B = Ax + (B - A)$.

Match: $A = 1, B - A = 0 \rightarrow B = 1$.

Integrate: $\int \left(\frac{1}{x-1} + \frac{1}{(x-1)^2} \right) dx = \ln |x - 1| - \frac{1}{x-1} + C$.

Example 19: Quadratic Factor

$$\int \frac{1}{x(x^2+1)} dx.$$

Template: $\frac{1}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+C}{x^2+1}$.

Clear: $1 = A(x^2 + 1) + (Bx + C)x = (A + B)x^2 + Cx + A$.

Match: $A + B = 0, C = 0, A = 1 \rightarrow B = -1$.

Integrate: $\int \left(\frac{1}{x} - \frac{x}{x^2+1} \right) dx = \ln |x| - \frac{1}{2} \ln(x^2 + 1) + C$.

Example 20: When Long Division Is Required First

$\int \frac{x^3}{x^2+1} dx$. Degree: $3 \geq 2 \rightarrow$ divide.

$x^3 \div (x^2 + 1): x^3 = x(x^2 + 1) - x$. So $\frac{x^3}{x^2+1} = x - \frac{x}{x^2+1}$.

Integrate: $\int x dx - \int \frac{x}{x^2+1} dx = \frac{x^2}{2} - \frac{1}{2} \ln(x^2 + 1) + C$.

Up to here: Parts (LIATE). Trig integrals (odd $\rightarrow$ peel+sub, even $\rightarrow$ half-angle, tan/sec $\rightarrow$ save sec²). Trig sub ($\sqrt{\text{form}}$ $\rightarrow$ sin/tan/sec). Partial fractions (factor $\rightarrow$ template $\rightarrow$ solve $\rightarrow$ integrate). All four techniques integrated into one decision tree.

Common Mistakes

Mistake 1: Choosing u and dv backwards in parts

Wrong: $u = e^x$, $dv = x dx$ for $\int x e^x dx$. Then $\int v du = \int \frac{x^2}{2} e^x dx$ — worse! **Right:** $u = x$, $dv = e^x dx$. The new integral $\int e^x dx$ is simpler.

Mistake 2: Using trig sub when u -sub is faster

Wrong: $\int \frac{x}{\sqrt{1-x^2}} dx \rightarrow$ trig sub leads to $\int \sin \theta d\theta$. **Right:** $u = 1 - x^2$, $du = -2x dx$. One step: $-\frac{1}{2} \int u^{-1/2} du = -\sqrt{1-x^2} + C$. Always check if a simple u -sub works first.

Mistake 3: Wrong trig substitution for the root form

Wrong: $x = 2 \sin \theta$ for $\sqrt{x^2 + 4}$. Then $\sqrt{4 \sin^2 \theta + 4}$ does NOT simplify. **Right:** $x = 2 \tan \theta$. Then $\sqrt{4 \tan^2 \theta + 4} = 2 \sec \theta$.

Mistake 4: Forgetting long division before partial fractions

Wrong: Decomposing $\frac{x^3}{x^2-1}$ directly. **Right:** Numerator degree (3) $\geq$ denominator degree (2). Divide first: result $= x + \frac{x}{x^2-1}$. Then decompose the remainder.

Mistake 5: Half-angle when an odd power is available

Wrong: Using $\sin^2 x = \frac{1-\cos 2x}{2}$ for $\int \sin^3 x \, dx$. **Right:** Peel one $\sin x$, convert $\sin^2 x = 1 - \cos^2 x$, u -sub with $u = \cos x$. Much faster.

What We Just Did

(1) Parts: $\int u \, dv = uv - \int v \, du$. LIATE for u . Cycle $\rightarrow$ apply twice, solve for I .

(2) Trig Integrals:

$\sin^m \cos^n$: odd $\rightarrow$ peel+convert+ u -sub. both even $\rightarrow$ half-angle.

$\tan^m \sec^n$: even $n \rightarrow$ save $\sec^2$, $u = \tan$. odd $m \rightarrow$ save $\sec \tan$, $u = \sec$.

(3) Trig Sub: $\sqrt{a^2 - x^2} \rightarrow x = a \sin \theta$. $\sqrt{a^2 + x^2} \rightarrow x = a \tan \theta$. $\sqrt{x^2 - a^2} \rightarrow x = a \sec \theta$.

Right triangle for back-substitution.

(4) Partial Fractions:

$\deg(P) \geq \deg(Q)? \rightarrow$ long divide first.

Factor Q . Write template. Solve constants. Integrate each term.

Practice 1

$\int x \sin x \, dx$. Run the parts algorithm. LIATE: $u = x$ (Algebraic), $dv = \sin x \, dx$.

$\rightarrow$ Solutions: [Solutions](#)

Practice 2

$\int \cos^3 x \, dx$. Odd power of cosine $\rightarrow$ peel one, convert $\cos^2$, u -sub with $u = \sin x$.

$\rightarrow$ Solutions: [Solutions](#)

Practice 3

$\int \frac{dx}{x^2+4}$. Arctan formula directly, OR trig sub $x = 2 \tan \theta$. Show both.

→ Solutions: [Solutions](#)

Practice 4

$\int \frac{x+1}{x^2-3x+2} dx$. Factor denominator: $(x-1)(x-2)$. Template: $\frac{A}{x-1} + \frac{B}{x-2}$. Solve, integrate.

→ Solutions: [Solutions](#)

Practice 5: Real Battle (Constructive)

A classmate sees $\int \frac{x}{\sqrt{1-x^2}} dx$ and says: "Trig sub! $\sqrt{1-x^2}$ means $x = \sin \theta$." (a) Is there a faster way? Solve it with u -sub. (b) Now solve $\int \frac{1}{\sqrt{1-x^2}} dx$ — was trig sub the right call this time? (c) Write a one-sentence rule: when $\sqrt{a^2 - x^2}$ appears, how do you choose between u -sub and trig sub?

→ Solutions: [Solutions](#)

Basic Algebra Drill — Advanced Integration (10 Problems)

Identify the technique from the decision tree. Execute the procedure.

D1. $\int x \cos x dx$. Parts: $u = x$, $dv = \cos x dx$.

D2. $\int \ln(2x) dx$. Parts: $u = \ln(2x)$, $dv = dx$.

D3. $\int \sin^2 x \, dx$. Trig integral (even $\rightarrow$ half-angle).

D4. $\int \tan x \sec^2 x \, dx$. Tan/sec: $n = 2$ (even) $\rightarrow u = \tan x$.

D5. $\int \frac{dx}{\sqrt{9-x^2}}$. Trig sub: $x = 3 \sin \theta$.

D6. $\int \frac{dx}{x^2+9}$. Arctan formula: $\frac{1}{3} \arctan(x/3) + C$.

D7. $\int \frac{1}{x^2-x} \, dx$. Partial fractions: factor $x(x-1)$.

D8. $\int x^2 e^x \, dx$. Parts twice: first $u = x^2$, then $u = x$.

D9. $\int \sin x \cos x \, dx$. Two methods: $u = \sin x$ OR $\sin 2x = 2 \sin x \cos x$. Verify answers match.

D10. $\int \frac{x}{\sqrt{1-x^2}} \, dx$. Decision: trig sub or u -sub? Choose the faster one.

Solutions: [Solutions](#)

Advanced Algebra Drill — Advanced Integration (10 Problems)

Multi-step. May combine techniques.

A1. $\int e^x \cos x \, dx$. Cycling parts — apply twice, solve for I .

A2. $\int \sin^4 x \, dx$. Repeated half-angle: $(\frac{1-\cos 2x}{2})^2$, expand, half-angle again.

A3. $\int \frac{dx}{(x^2+1)^2}$. Trig sub $x = \tan \theta$. Right triangle for back-sub.

A4. $\int \frac{x^3}{\sqrt{x^2+1}} \, dx$. $u = x^2 + 1$ is faster than trig sub — try it.

A5. $\int \frac{x^2+2x-1}{(x-1)(x^2+1)} \, dx$. Quadratic factor: template $\frac{A}{x-1} + \frac{Bx+C}{x^2+1}$.

A6. $\int \arctan x \, dx$. Parts: $u = \arctan x$, $dv = dx$. Result needs u -sub.

A7. $\int \sec^3 x \, dx$. Parts: $u = \sec x$, $dv = \sec^2 x \, dx$. Classic — it cycles.

A8. $\int \frac{\sqrt{x^2-4}}{x} dx$. Trig sub $x = 2 \sec \theta$. Right triangle needed.

A9. $\int x \arcsin x dx$. Parts: $u = \arcsin x$, $dv = x dx$. Result needs trig sub or clever u -sub.

A10. Derive the reduction formula: $\int \sin^n x dx = -\frac{1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} \int \sin^{n-2} x dx$.
(Parts: $u = \sin^{n-1} x$, $dv = \sin x dx$.)

Solutions: [Solutions](#)

Today's Procedure

When you see...	Do this...
Product of unrelated functions	Parts: LIATE for u , $dv = \text{rest}$. Cycle $\rightarrow$ twice, solve for I
$\sin^m x \cos^n x$, m or n odd	Trig integral: peel one, convert rest, u -sub
$\sin^m x \cos^n x$, both even	Half-angle: reduce powers, integrate, repeat if needed
$\tan^m x \sec^n x$, n even	Save $\sec^2 x$, $u = \tan x$
$\sqrt{a^2 - x^2}$	Trig sub: $x = a \sin \theta$
$\sqrt{a^2 + x^2}$	Trig sub: $x = a \tan \theta$
$\sqrt{x^2 - a^2}$	Trig sub: $x = a \sec \theta$
$\frac{P(x)}{Q(x)}$, Q factorable	Partial fractions: factor $\rightarrow$ template $\rightarrow$ solve $\rightarrow$ integrate
$\frac{P(x)}{Q(x)}$, $\deg P \geq \deg Q$	Long divide first, then partial fractions on remainder

Terminology

What we call it	Math term	Notation
reverse product rule	integration by parts	$\int u dv = uv - \int v du$

What we call it	Math term	Notation
priority order for u	LIATE	Log, Inverse trig, Algebraic, Trig, Exponential
peel-convert-substitute	odd-power trig strategy	$\sin^3 x = \sin x(1 - \cos^2 x)$
halve the angle	half-angle formula	$\sin^2 x = \frac{1 - \cos 2x}{2}$
remove root via trig	trigonometric substitution	$x = a \sin \theta$, etc.
triangle for back-sub	reference triangle	SOH CAH TOA
split rational function	partial fraction decomposition	$\frac{A}{x-a} + \frac{B}{x-b} + \dots$
numerator too big	improper rational function	do long division first
integral appears on both sides	cycling / recursive integral	solve for I algebraically